

# FIRE SAFETY AUTOMATION SYSTEM

*Using Arduino and Blynk*

---

An IoT-based Gas Leak & Overheat Protection System

**Aditya Verma**

Roll No. 24CE10026

Indian Institute of Technology Kharagpur

## Project Description

This IoT-based fire and air quality safety system is designed to detect gas leaks, monitor ambient temperature, and automatically control appliances such as fans, alarms, and water pumps. The system is powered by an Arduino with an Ethernet Shield and is integrated with the Blynk IoT Cloud for real-time monitoring, alerts, and remote tracking.

### Components Used

| #  | Component              |
|----|------------------------|
| 1  | Arduino Uno            |
| 2  | Ethernet Shield        |
| 3  | MQ135 Gas Sensor       |
| 4  | DHT11 Sensor           |
| 5  | Buzzer                 |
| 6  | Jumper Wires           |
| 7  | Breadboard             |
| 8  | 2-Channel Relay Module |
| 9  | Mini Water Pump        |
| 10 | 12V Fan                |
| 11 | Power Supply           |

## Sensor Technology

### MQ135 Gas Sensor

MQ135 is a semiconductor-based gas sensor used to monitor the presence of harmful gases and smoke in the atmosphere. It works on the principle that the electrical resistance of the sensing material changes in the presence of gases, and the output electrical signal given by the MQ135 changes accordingly.

The core sensing element of the sensor is made of tin dioxide ( $\text{SnO}_2$ ). It has low conductivity in clean air, and on coming into contact with toxic gases, its conductivity increases.

### DHT11 Sensor

The DHT11 sensor is used to measure temperature and humidity.

For measuring temperature, it uses a thermistor with a negative temperature coefficient. As the temperature increases, the resistance decreases, and the corresponding change in temperature is calculated accordingly.

For measuring humidity, it uses a capacitive humidity sensor. This component consists of two electrodes with a moisture-absorbing polymer substrate between them. Changes in humidity affect the conductivity and capacitance of the substrate, which is measured and converted into an electrical signal by the internal microcontroller.

## Working of the Fire Safety Automation System

---

### Gas Detection

1. Uses an analog gas sensor (MQ135) to detect hazardous gas levels.
2. Triggers an alarm and activates a fan when gas exceeds a safe threshold.
3. Sends an SOS alert to the Blynk app via push notification.

### Temperature Monitoring

4. Employs a DHT11 sensor to measure the surrounding temperature.
5. Automatically turns on a water pump if temperature exceeds 40°C (overheat protection).
6. Sends temperature data to Blynk for remote monitoring.

### Blynk Integration

- Real-time gas and temperature readings on a mobile dashboard.
- Push notifications for critical events like gas leakage or overheating.
- Scalable template with virtual pins (V0 for gas, V1 for temperature).

### Automatic Actuation

- Relay modules control fan, alarm, and water pump based on sensor thresholds.
- Fully autonomous operation, with optional manual overrides.

## Circuit Diagram

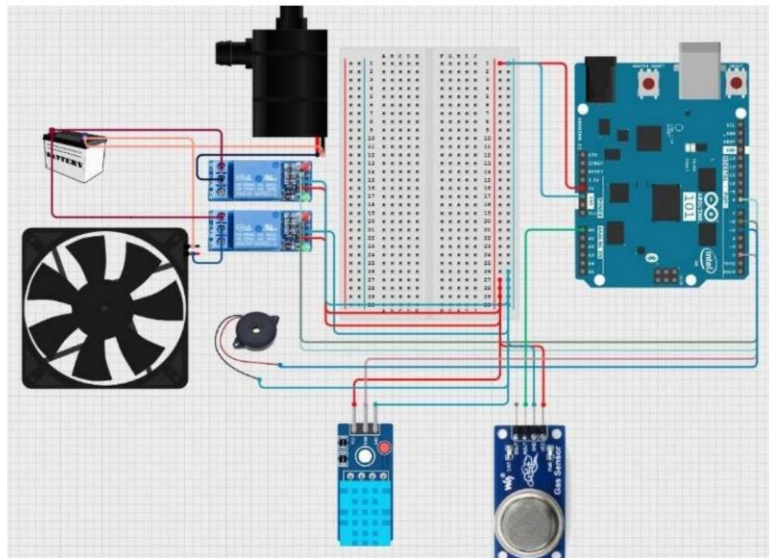


Figure 1: Circuit diagram showing Arduino, sensors, relays, and actuators

## Blynk Interface & Alerts

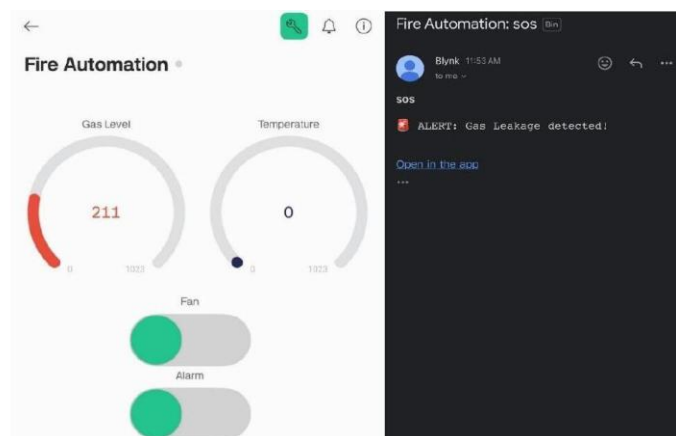
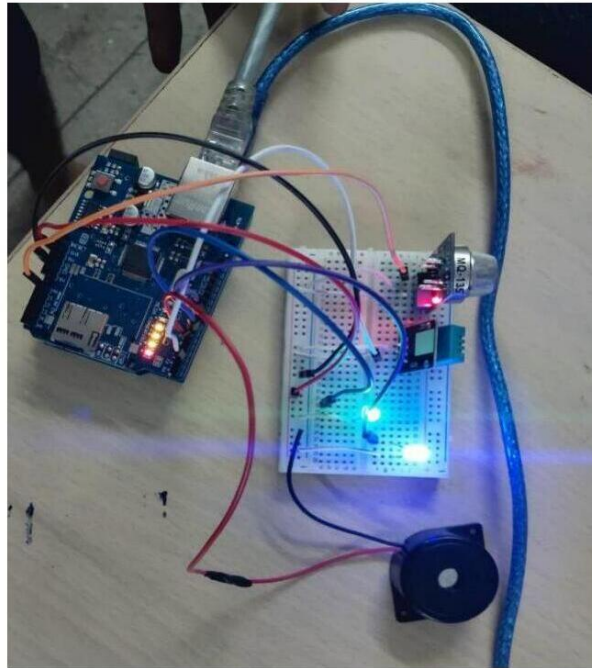


Figure 2: Blynk mobile dashboard (left) and gas leakage alert notification via email (right)

## Project Photograph



*Figure 3: Assembled hardware prototype*

## Program Code

The complete Arduino sketch used to drive the system, including Blynk virtual pin handlers and the main sensing/actuation loop, is given below.

```
#define BLYNK_TEMPLATE_ID "TMPL3HSsdzzSr"
#define BLYNK_TEMPLATE_NAME "Fire Automation"
#define BLYNK_AUTH_TOKEN "NEvbKuBmHlp29CmCa0Ge-id27W7Q0J1b"

#include <SPI.h>
#include <Ethernet.h>
#include <BlynkSimpleEthernet.h>
#include <DHT.h>

const int gasSensorPin = A0;

// DHT11 setup
#define DHTPIN 2
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);

// Blynk virtual pins
#define GAS_VPIN V0
#define TEMP_VPIN V1
#define FAN_VPIN V2
#define ALARM_VPIN V3
#define WATERPUMP_VPIN V4

const int alarm_pin = 5;
const int fan_pin = 6;
const int waterPump_pin = 8;

void setup() {
  Serial.begin(9600);
  Blynk.begin(BLYNK_AUTH_TOKEN);
  dht.begin();
  pinMode(alarm_pin, OUTPUT);
  pinMode(fan_pin, OUTPUT);
  pinMode(waterPump_pin, OUTPUT);
}

BLYNK_WRITE(FAN_VPIN) {
  int fanState = param.asInt(); // 1 = ON, 0 = OFF
  digitalWrite(fan_pin, fanState);
}

BLYNK_WRITE(ALARM_VPIN) {
  int alarmState = param.asInt(); // 1 = ON, 0 = OFF
  digitalWrite(alarm_pin, alarmState);
}

BLYNK_WRITE(WATERPUMP_VPIN) {
  int waterpumpState = param.asInt(); // 1 = ON, 0 = OFF
  digitalWrite(waterPump_pin, waterpumpState);
}
```

```
void loop() {
  Blynk.run();

  // GAS SENSOR
  int gasValue = analogRead(gasSensorPin);
  Serial.print("Gas Value: ");
  Serial.println(gasValue);
  Blynk.virtualWrite(GAS_VPIN, gasValue);

  if (gasValue > 400) {
    digitalWrite(alarm_pin, HIGH);
    digitalWrite(fan_pin, HIGH);
    Blynk.logEvent("sos", "ALERT: Gas Leakage detected!");
    Blynk.virtualWrite(FAN_VPIN, 1);
    Blynk.virtualWrite(ALARM_VPIN, 1);
  }

  // DHT11 SENSOR
  float temperature = dht.readTemperature();
  if (!isnan(temperature)) {
    Serial.print("Temperature: ");
    Serial.print(temperature);
    Serial.println(" C");
    Blynk.virtualWrite(TEMP_VPIN, temperature);

    if (temperature > 40) {
      digitalWrite(waterPump_pin, HIGH);
      digitalWrite(alarm_pin, HIGH);
      Blynk.logEvent("overheat", "Overheat! Water pump ON");
      Blynk.virtualWrite(WATERPUMP_VPIN, 1);
      Blynk.virtualWrite(ALARM_VPIN, 1);
    }
  } else {
    Serial.println("Failed to read from DHT11 sensor.");
  }

  delay(1000);
}
```

## Applications in Civil Engineering

---

### Commercial and Industrial Facilities

Factories, warehouses, and office buildings benefit from automated fire suppression systems that target specific areas to minimize damage.

### Smart Cities

Urban planning projects can use city-wide Building Automation Systems (BAS) to connect fire safety measures with municipal emergency services.

### Construction Sites

Fire risk assessment software integrated with Building Information Modelling (BIM) helps identify potential hazards during the design phase.

### Transportation Infrastructure

Tunnels, airports, and train stations require advanced fire suppression systems to manage risks in confined spaces.